

Contoso Hotel Group - Analytics Dashboard

AI-Augmented Business Intelligence & Machine Learning Analytics

11 Pages | 3 Machine Learning Models | 45K Customers | \$63.98M Revenue

Key Highlights

- \$63.98M Revenue Analyzed
- 3 ML Models Embedded
- AI Anomaly Detection & Forecasting
- Key Influencers AI Visual
- 68% Prediction Accuracy

Tools & Technologies

- Power BI Desktop & DAX - Python
- Scikit-learn, pandas, matplotlib
- Machine Learning
 - Random Forest
 - K-Means
 - Logistic Regression
- Star Schema Data Modelling

Developed by:

Jagminder Singh
Master of Data Science and Decisions (UNSW) |
2025

Executive Summary - 2025

Country

All



Hotel

All



Month Year

All



Activity

All



Reset

Home

Executive Dashboard

Revenue Summary

Revenue Growth

Revenue Decomposition

Reservations

Customer Activity Analysis

Leisure Revenue Analysis

Key Performing Indicators

Customer Segmentation K-Means - ML

Revenue Drivers (Random Forest) —...

Return Prediction - Logistic Regression

\$63.98M

Gross Revenue

45K

Total Customers

11.70

Return Rate

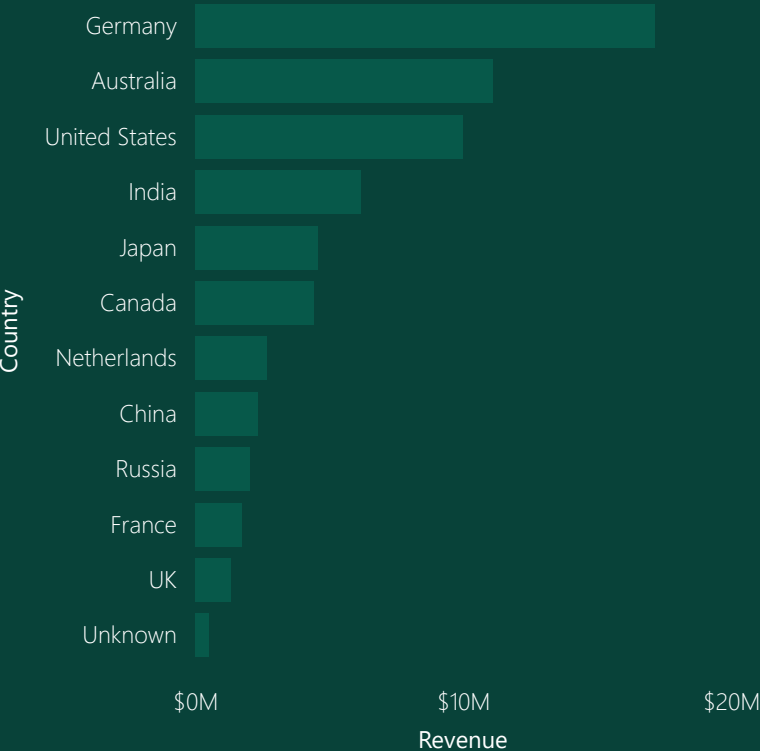
1K

Sales per reservation

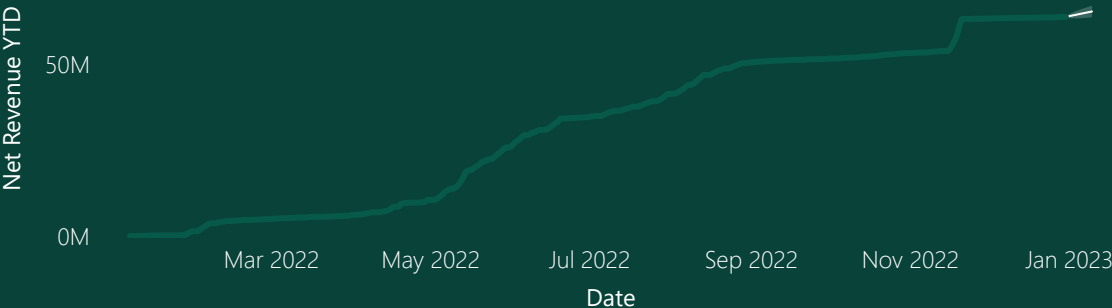
Germany

Top Country

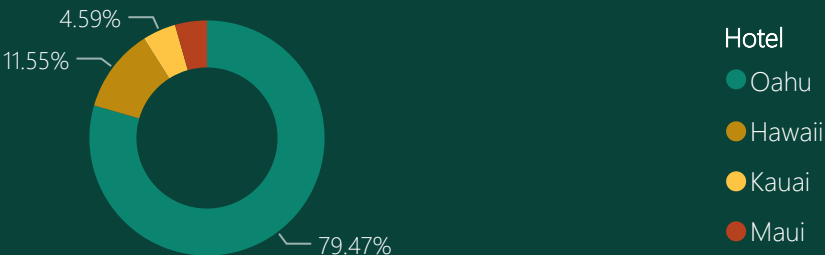
Revenue by Country



Revenue Growth Year to Date



Revenue by Hotel



Revenue Summary - 2025

Country

All



Hotel

All



Month Year

All



Activity

All



Reset

Home

Executive Dashboard

Revenue Summary

Revenue Growth

Revenue Decomposition

Reservations

Customer Activity Analysis

Leisure Revenue Analysis

Key Performing Indicators

Customer Segmentation K-Means - ML

Revenue Drivers (Random Forest) —...

Return Prediction - Logistic Regression

\$63.98M

Gross Revenue

45K

Total Customers

40K

New Customers

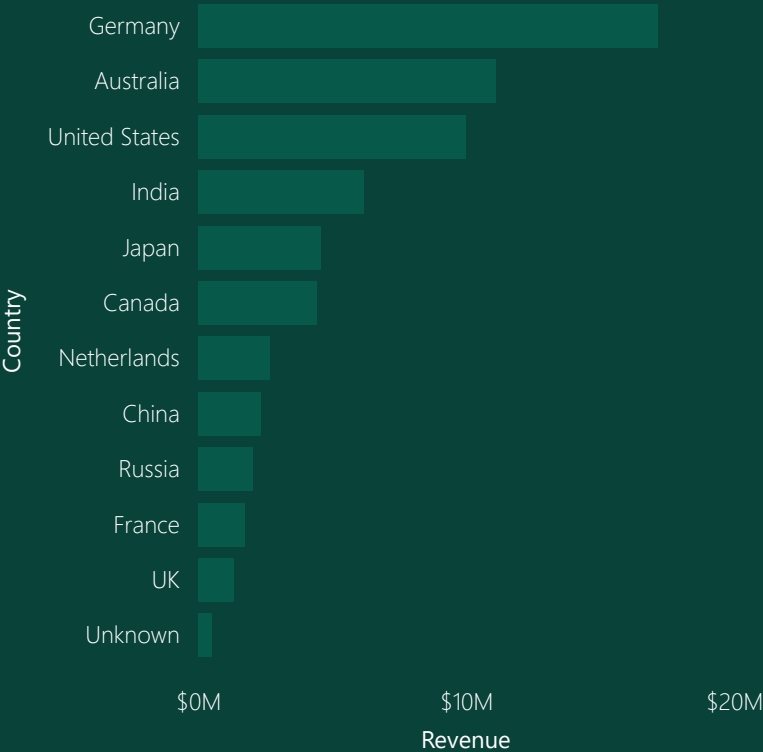
5K

Repeat Customers

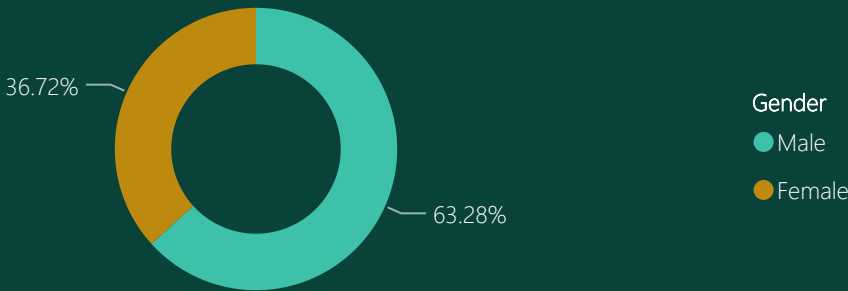
11.70

Return Rate

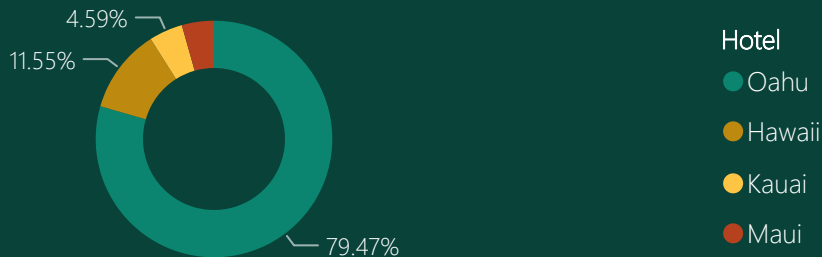
Revenue by Country



Revenue by Gender



Revenue by Hotel



Revenue Growth - 2025

Country

All



Hotel

All



Month Year

All



Activity

All



Reset

Home

Executive Dashboard

Revenue Summary

Revenue Growth

Revenue Decomposition

Reservations

Customer Activity Analysis

Leisure Revenue Analysis

Key Performing Indicators

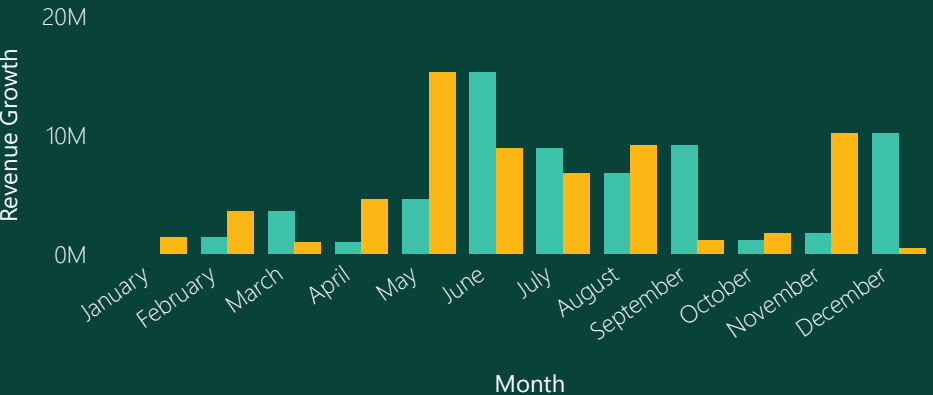
Customer Segmentation K-Means - ML

Revenue Drivers (Random Forest) —...

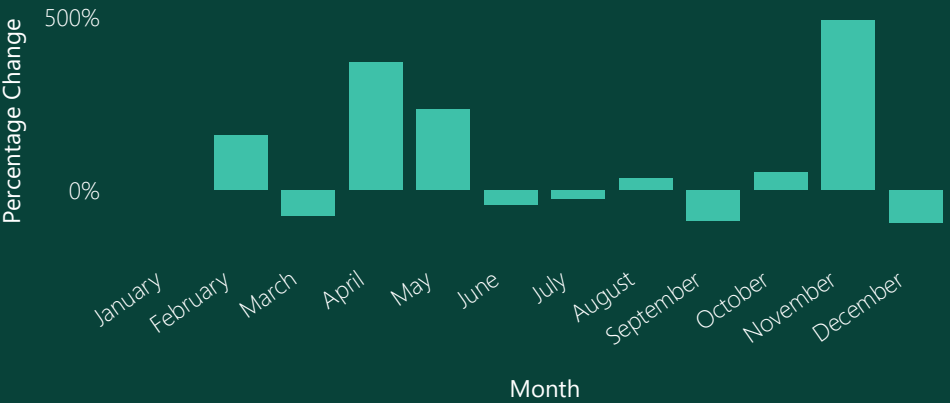
Return Prediction - Logistic Regression

MOM Revenue Growth

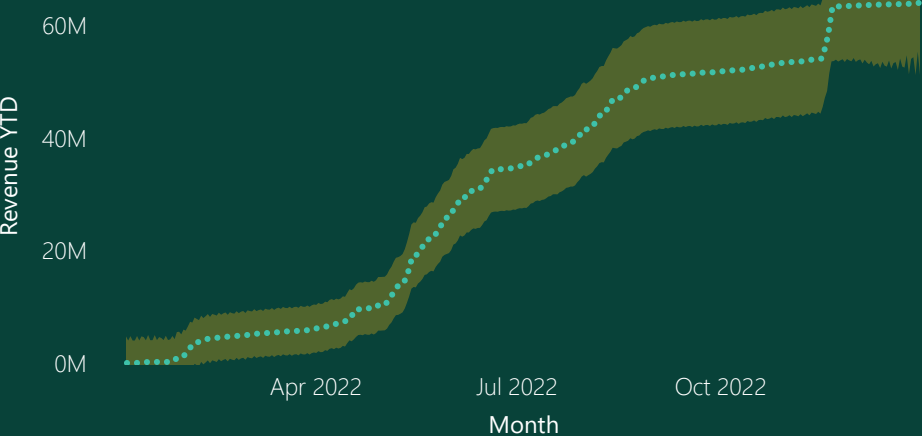
Previous Month Revenue Current Month Revenue



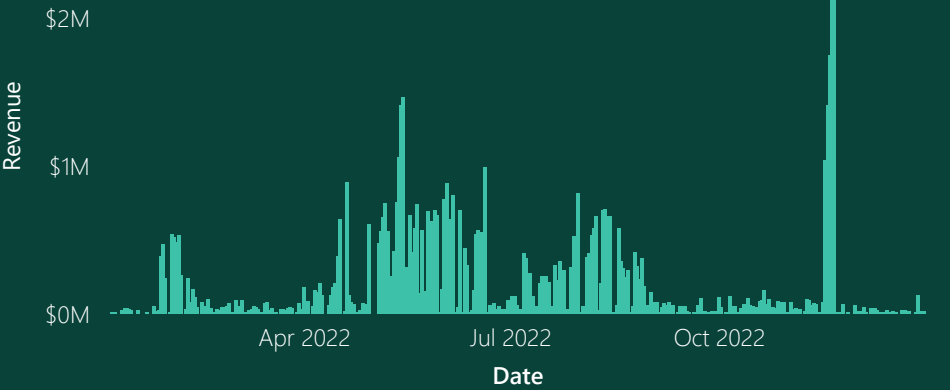
Revenue Growth Percentage Change



YTD Revenue Growth



Revenue Share by Month



Revenue Decomposition

Country

All

Hotel

All

Month Year

All

Activity

All

Reset

Home

Executive Dashboard

Revenue Summary

Revenue Growth

Revenue Decomposition

Reservations

Customer Activity Analysis

Leisure Revenue Analysis

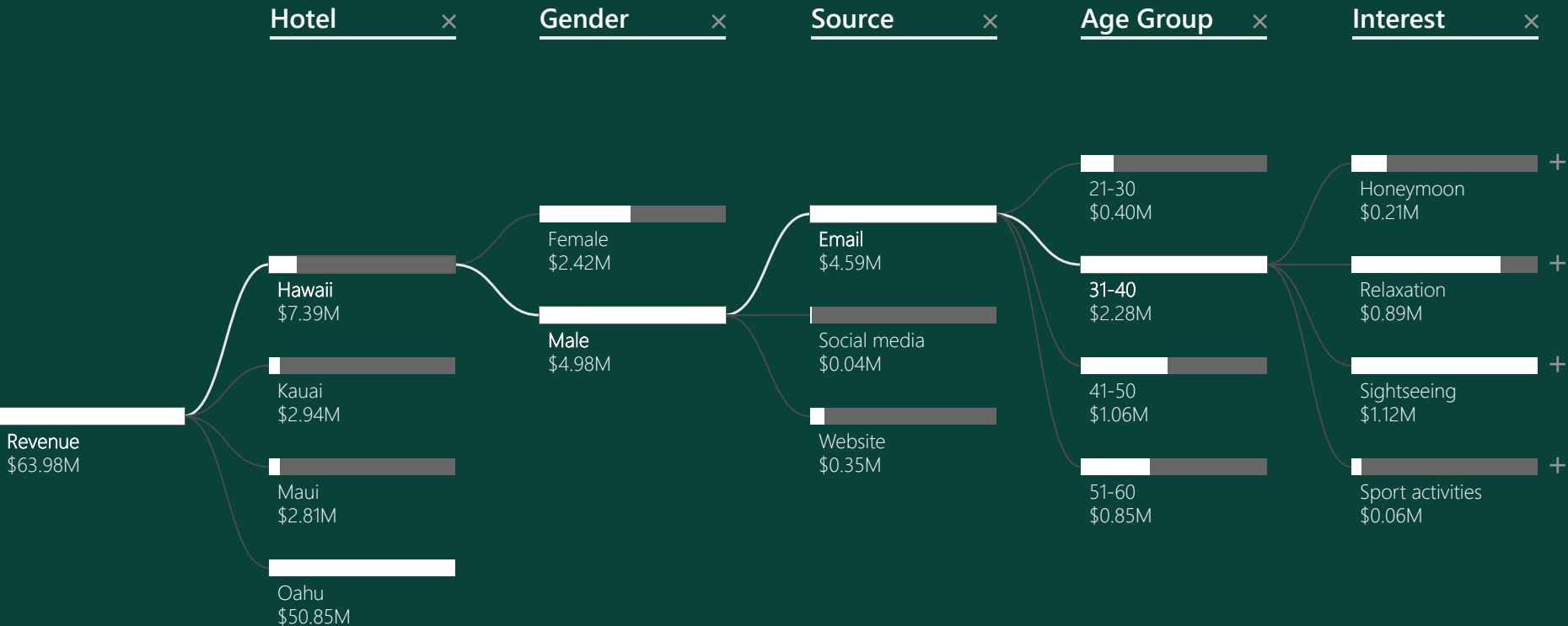
Key Performing Indicators

Customer Segmentation K-Means - ML

Revenue Drivers (Random Forest) —...

Return Prediction - Logistic Regression

Revenue Decomposition Analysis



Reservations

Country

All



Hotel

All



Month Year

All



Activity

All



Reset

Home

Executive Dashboard

Revenue Summary

Revenue Growth

Revenue Decomposition

Reservations

Customer Activity Analysis

Leisure Revenue Analysis

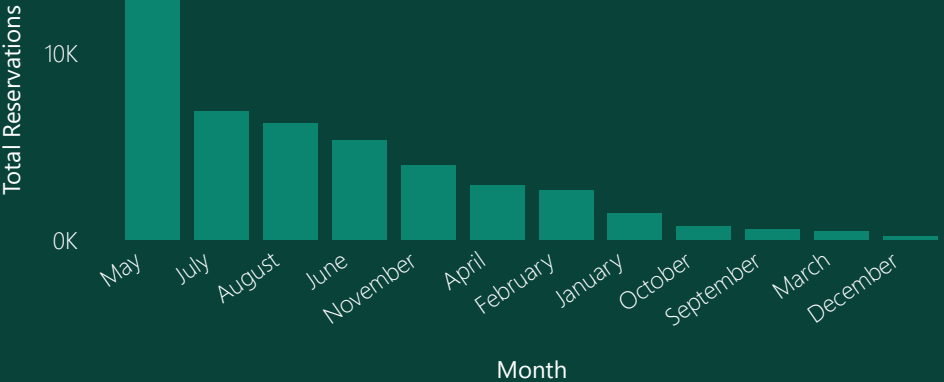
Key Performing Indicators

Customer Segmentation K-Means - ML

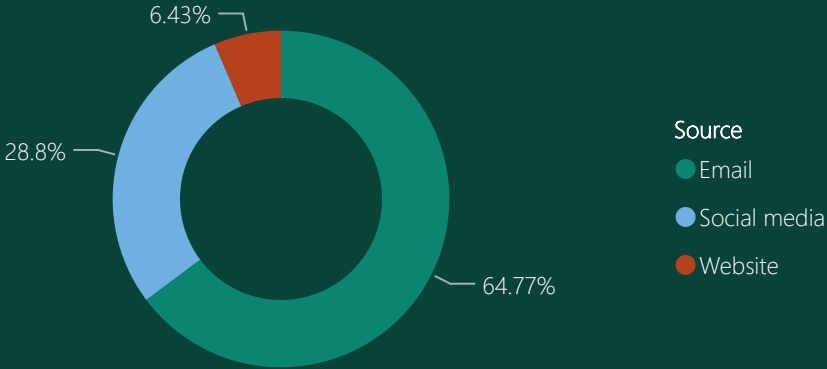
Revenue Drivers (Random Forest) —...

Return Prediction - Logistic Regression

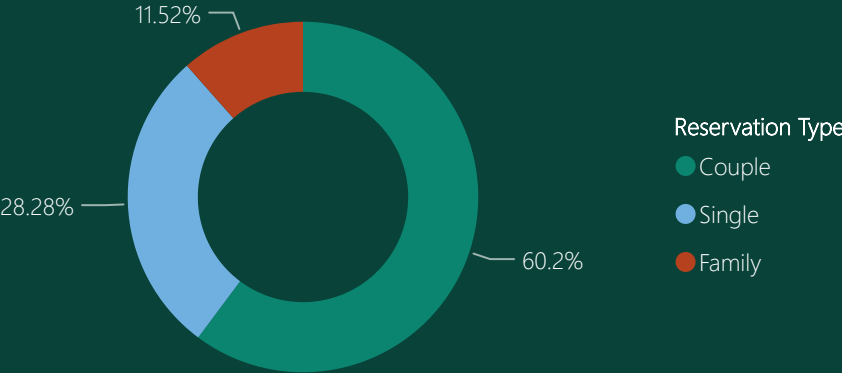
Total Reservations By Month



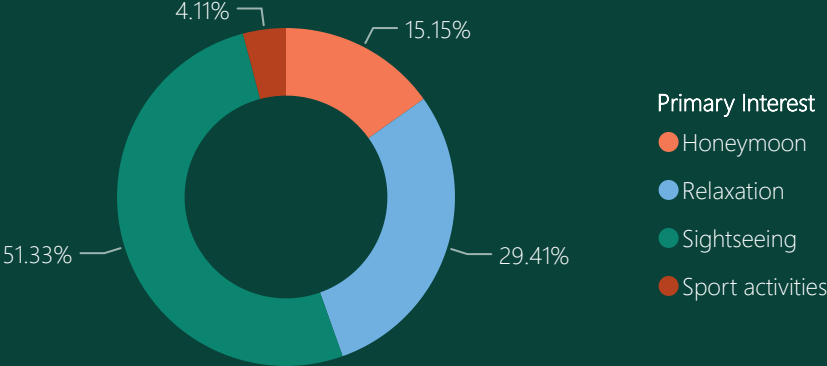
Total Reservations by Source



Total Reservations by Reservation Type



Total Reservations by Primary Interest



Customer Activity Analysis

Country

All



Hotel

All



Month Year

All



Activity

All



Reset

Home

Executive Dashboard

Revenue Summary

Revenue Growth

Revenue Decomposition

Reservations

Customer Activity Analysis

Leisure Revenue Analysis

Key Performing Indicators

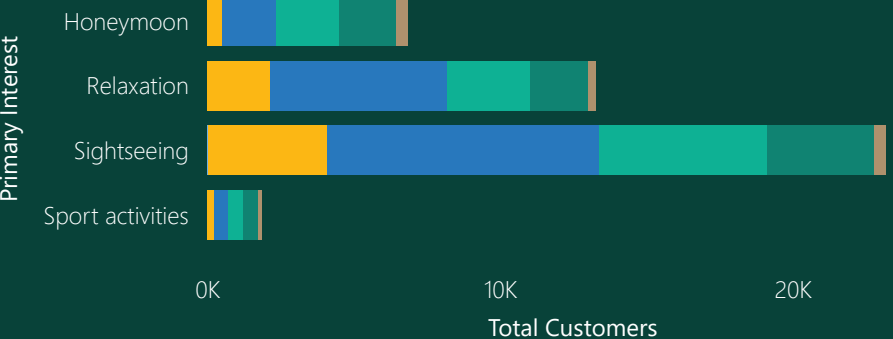
Customer Segmentation K-Means - ML

Revenue Drivers (Random Forest) —...

Return Prediction - Logistic Regression

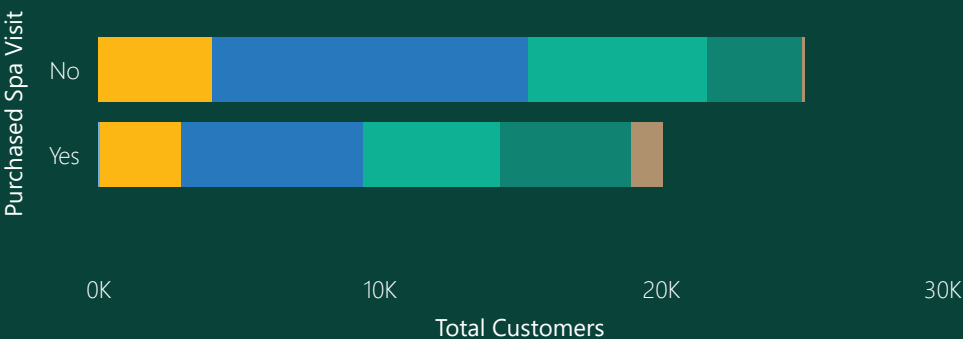
Total Customers by Primary Interest and Age Group

Age Group <21 21-30 31-40 41-50 51-60 >60



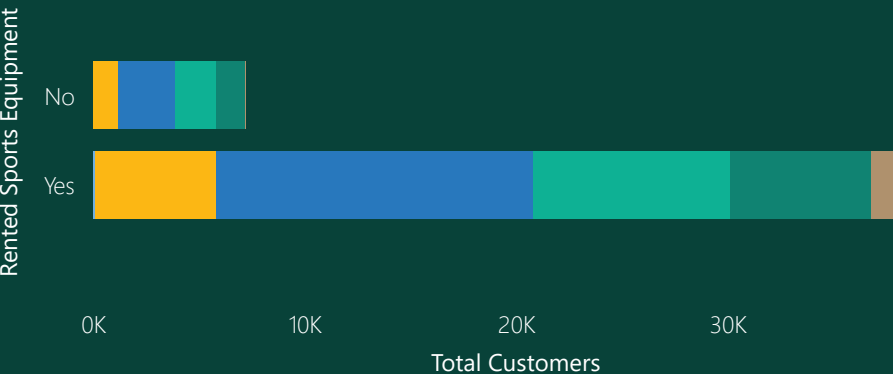
Total Customers by Purchased Spa Visit and Age Group

Age Group <21 21-30 31-40 41-50 51-60 >60



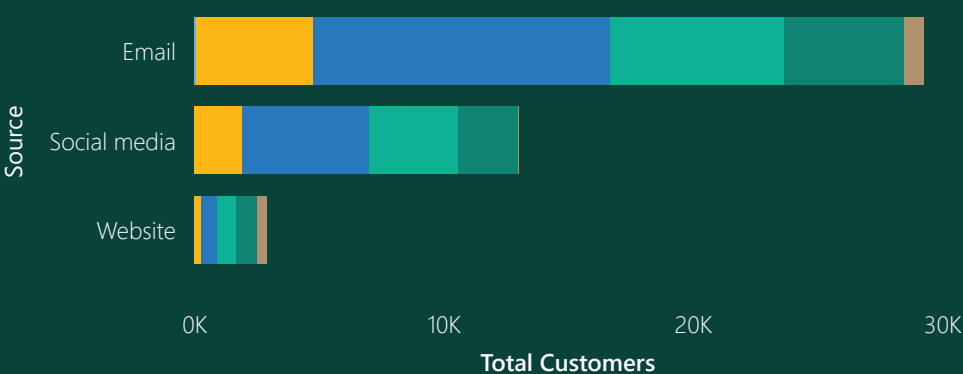
Total Customers by Rented Sports Equipment and Age Group

Age Group <21 21-30 31-40 41-50 51-60 >60



Total Customers by Source and Age Group

Age Group <21 21-30 31-40 41-50 51-60 >60



Leisure Revenue Analysis

Country

All



Hotel

All



Month Year

All



Activity

All



Reset

Home

Executive Dashboard

Revenue Summary

Revenue Growth

Revenue Decomposition

Reservations

Customer Activity Analysis

Leisure Revenue Analysis

Key Performing Indicators

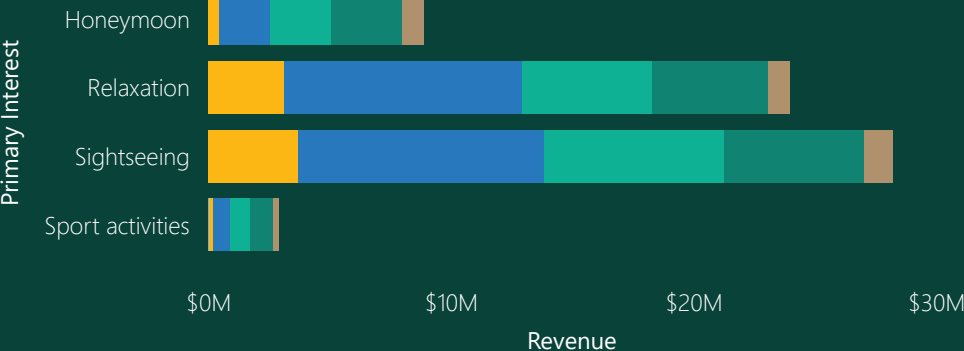
Customer Segmentation K-Means - ML

Revenue Drivers (Random Forest) —...

Return Prediction - Logistic Regression

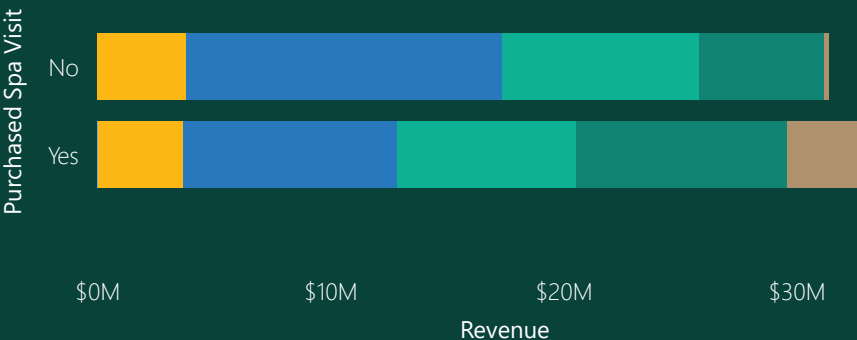
Revenue by Primary Interest and Age Group

Age Group <21 21-30 31-40 41-50 51-60 >60



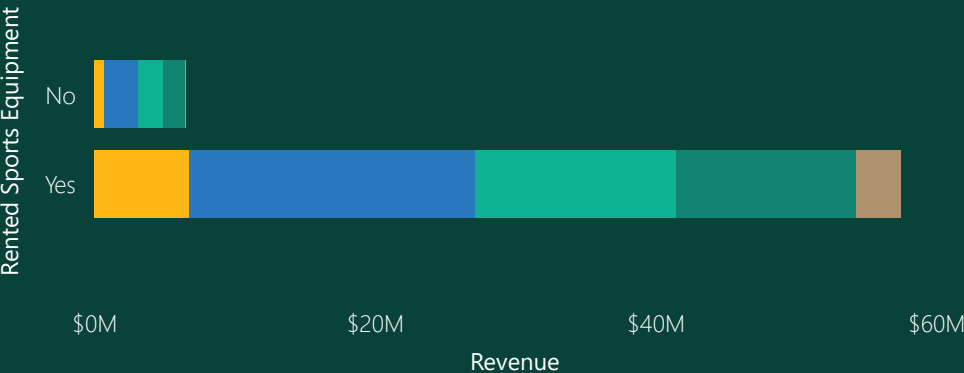
Revenue by Purchased Spa Visit and Age Group

Age Group <21 21-30 31-40 41-50 51-60 >60



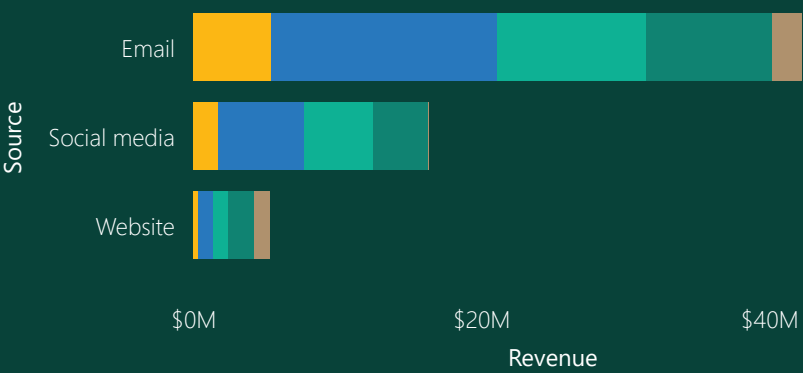
Revenue by Rented Sports Equipment and Age Group

Age Group <21 21-30 31-40 41-50 51-60 >60



Revenue by Source and Age Group

Age Group <21 21-30 31-40 41-50 51-60 >60



Key Performing Indicators

Country

All

Hotel

All

Month Year

All

Activity

All

Reset

Home

Executive Dashboard

Revenue Summary

Revenue Growth

Revenue Decomposition

Reservations

Customer Activity Analysis

Leisure Revenue Analysis

Key Performing Indicators

Customer Segmentation K-Means - ML

Revenue Drivers (Random Forest) —...

Return Prediction - Logistic Regression

Key influencers Top segments

What influences Return Rate % to Increase ?

When...

....the average of Return Rate % increases by

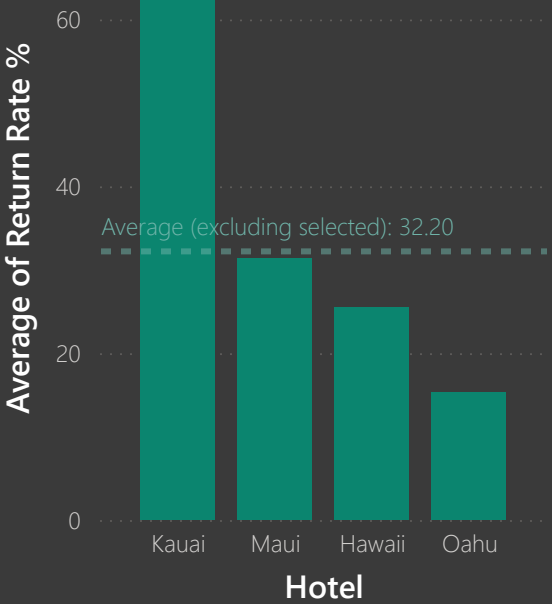
Hotel is Kauai

40.31

Gender is Female

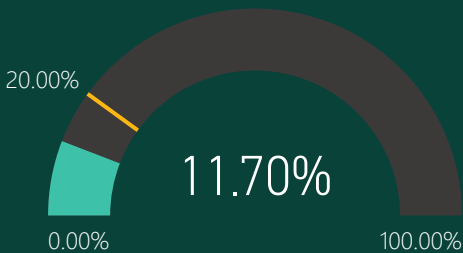
24.02

← Return Rate % is more likely to increase when Hotel is Kauai than otherwise (on average).

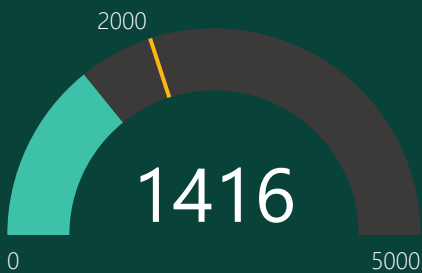


☐ Only show values that are influencers

Customer Return Rate



Sales per reservation



Customer Visits



Customer Segmentation

Country

All



Hotel

All



Month Year

All



Activity

All



Reset

Home

Executive Dashboard

Revenue Summary

Revenue Growth

Revenue Decomposition

Reservations

Customer Activity Analysis

Leisure Revenue Analysis

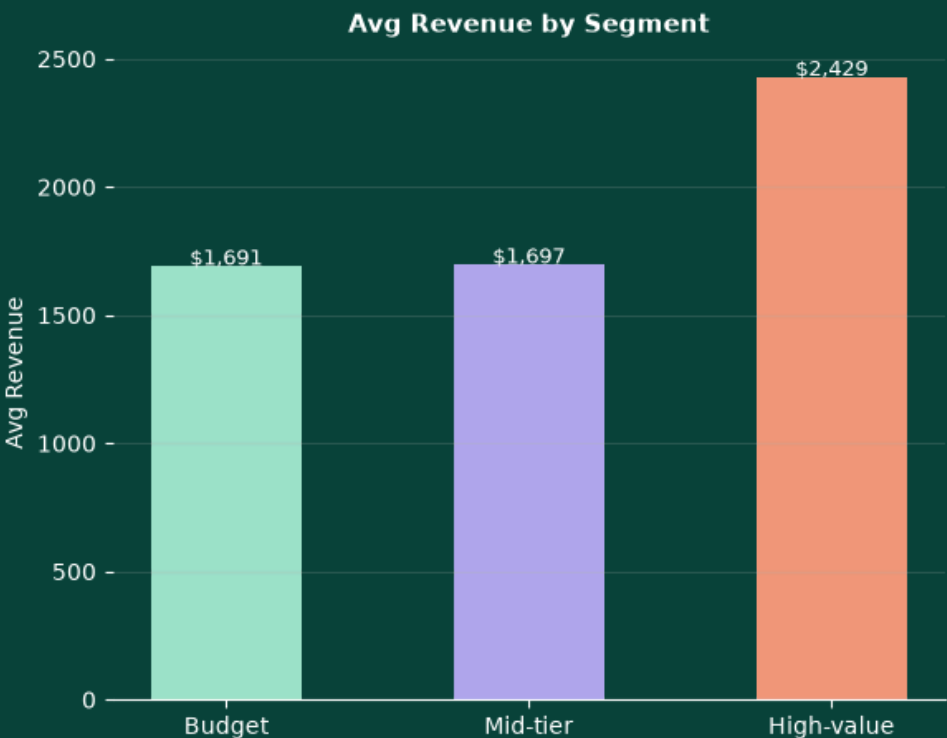
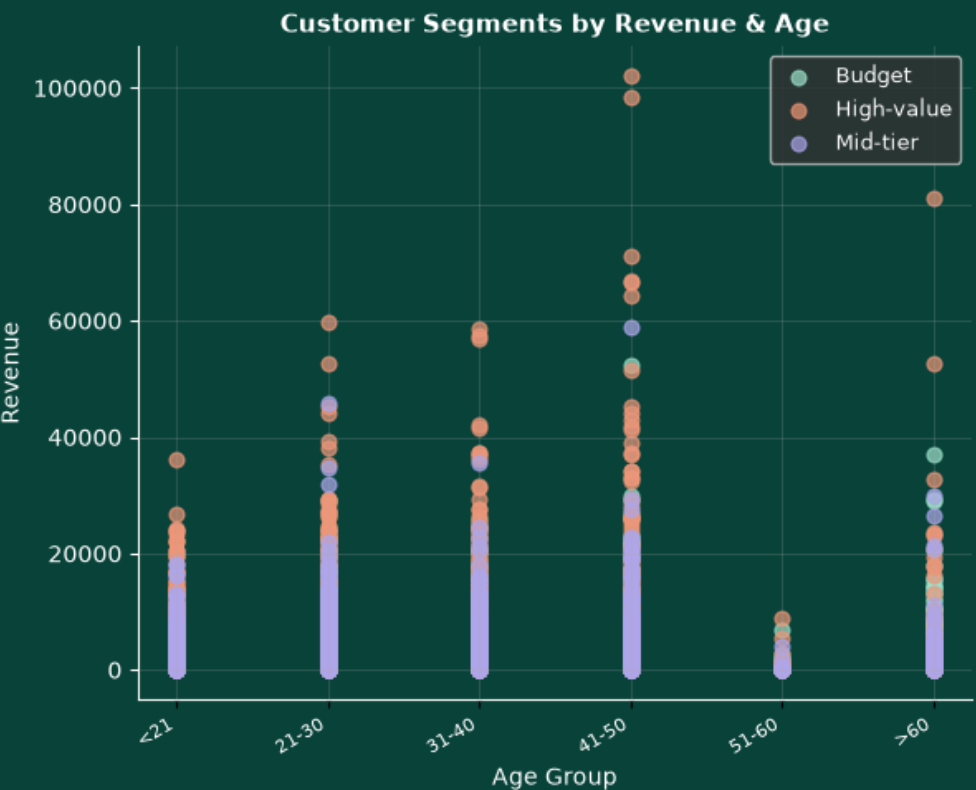
Key Performing Indicators

Customer Segmentation K-Means - ML

Revenue Drivers (Random Forest) —...

Return Prediction - Logistic Regression

Customer Segmentation (K-Means) — Machine Learning Analysis



Revenue Drivers

Country

All



Hotel

All



Month Year

All



Activity

All



Reset

Home

Executive Dashboard

Revenue Summary

Revenue Growth

Revenue Decomposition

Reservations

Customer Activity Analysis

Leisure Revenue Analysis

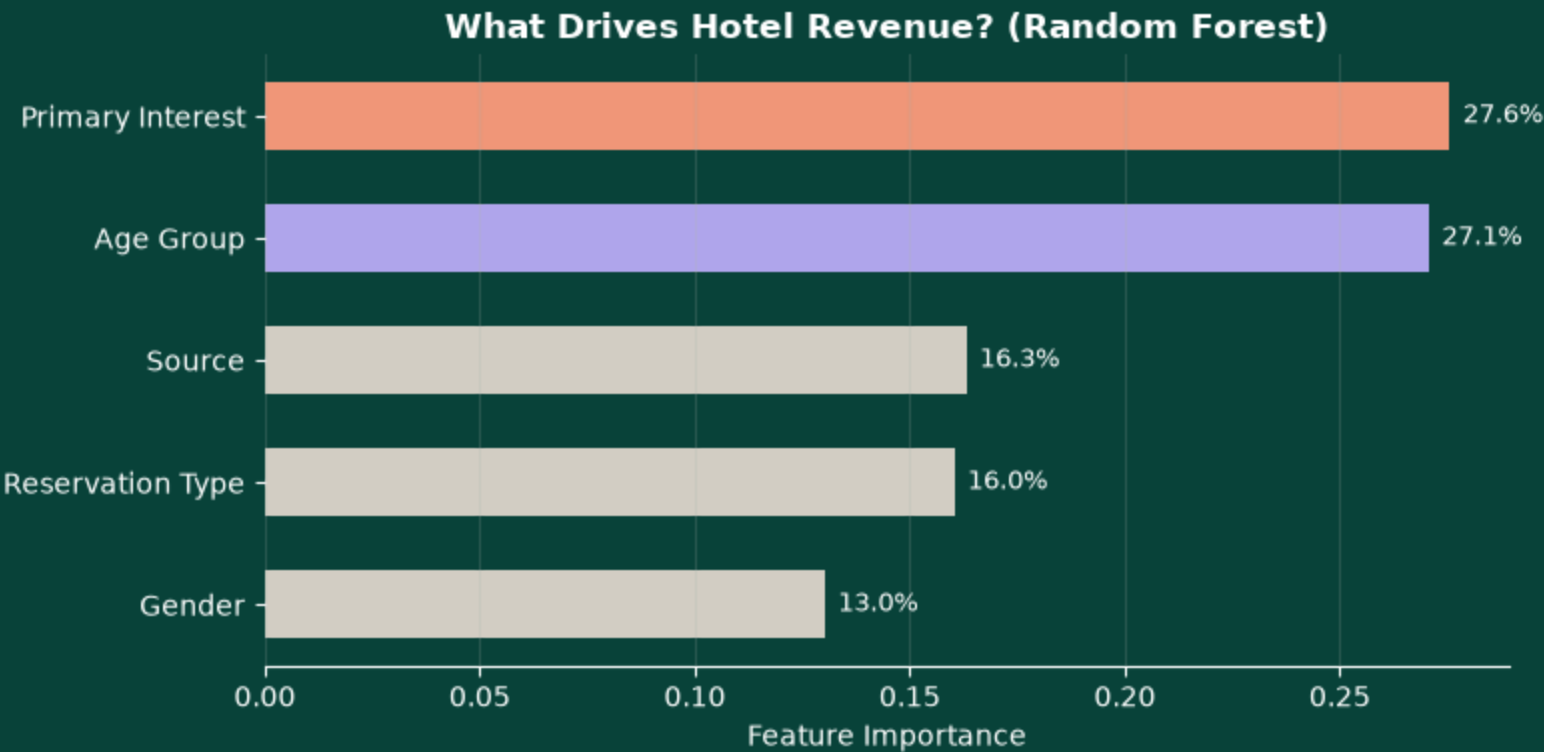
Key Performing Indicators

Customer Segmentation K-Means - ML

Revenue Drivers (Random Forest) —...

Return Prediction - Logistic Regression

Revenue Drivers (Random Forest) — Machine Learning Analysis



Return Prediction

Country

All



Hotel

All



Month Year

All



Activity

All



Reset

Home

Executive Dashboard

Revenue Summary

Revenue Growth

Revenue Decomposition

Reservations

Customer Activity Analysis

Leisure Revenue Analysis

Key Performing Indicators

Customer Segmentation K-Means - ML

Revenue Drivers (Random Forest) —...

Return Prediction - Logistic Regression

Customer Return Prediction - Logistic Regression

